LAB 2

Agile Backlog Creation & Sprint Simulation in Jira

PROJECT

Public Bus Live Tracking & Crowding Estimator

JIRA PROJECT KEY

PBLT

STUDENT

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Prepared using Jira Software (Company-managed Scrum project)

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1 Objective

This lab applies Agile/Scrum practices to the **Public Bus Live Tracking & Crowding Estimator** project using Jira Software. The functional requirements identified in Lab 1 were decomposed into a structured backlog, estimated, and executed across two simulated sprints so that the resulting workflow and progress data could be tracked and analysed directly inside Jira.

The exercise follows the standard Agile decomposition pipeline:

1. Functional Requirements
2. Epics
3. User Stories
4. Priorities
5. Story Points
6. Sprint Planning
7. Sprint Execution
8. Burndown Analysis

Each stage was carried out inside a company-managed Jira Scrum project (key **PBLT**): Epics group related functionality, User Stories express end-user value in the *As a / I want / So that* format, Story Points (Fibonacci scale) estimate relative effort, and the Scrum board tracks status transitions from **To Do** to **In Progress** to **Done** across Sprint 1 and Sprint 2.

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Project Overview

Project: Public Bus Live Tracking & Crowding Estimator

Jira Key: PBLT

The system is designed to give commuters real-time visibility into public bus locations, arrival estimates, and expected crowding, while giving operations staff tools to monitor fleet health and respond to unreliable data feeds. The scope covers:

- Real-time bus location tracking
- ETA (estimated time of arrival) calculation
- Bus crowding estimation
- Fleet monitoring
- Transit alerts for commuters
- Handling of stale or missing GPS data

Epic Overview

Epic	Purpose
Live Bus Tracking & ETA	Ingest live GPS data and surface accurate, real-time arrival estimates to riders.
Crowding Estimation	Estimate how crowded an approaching bus is likely to be, to help riders plan boarding.
Fleet Monitoring & Reliability	Give operations staff fleet-wide status visibility, alerting, and graceful handling of unreliable data.

3

Jira Project and Epics

The PBLT project was set up as a company-managed Scrum project in Jira. The three Epics below were created to group the backlog into major functional themes, matching the Epic panel shown in the Jira backlog view.

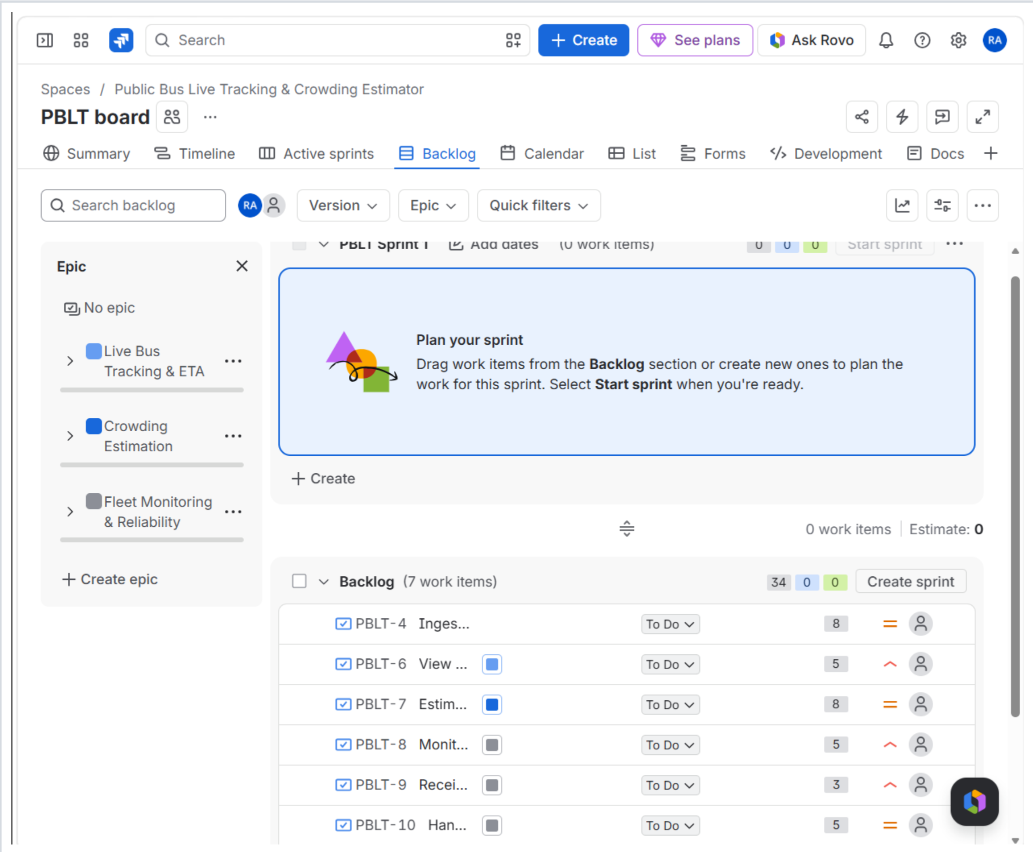

Figure 1. *Jira Backlog view showing the three project Epics (left panel) and the full backlog of six User Stories with their assigned story points, prior to Sprint 1 being started.*

1. Live Bus Tracking & ETA — Covers GPS ingestion and rider-facing ETA visibility.

2. Crowding Estimation — Covers estimation of bus occupancy/crowding levels.

3. Fleet Monitoring & Reliability — Covers fleet status monitoring, alerts, and stale-data handling.

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User Stories

Six User Stories were created under the three Epics above, each scoped to a single piece of user-facing or operational value.

ID	User Story	Epic	Priority	Points
PBLT-4	Ingest GPS and Calculate ETA	Live Bus Tracking & ETA	Medium	8
PBLT-6	View Live Bus ETA	Live Bus Tracking & ETA	High	5
PBLT-7	Estimate Bus Crowding	Crowding Estimation	Medium	8
PBLT-8	Monitor Fleet Status	Fleet Monitoring & Reliability	High	5
PBLT-9	Receive Transit Alerts	Fleet Monitoring & Reliability	High	3
PBLT-10	Handle Stale Data	Fleet Monitoring & Reliability	Medium	5

User Story Format

Each story was written in Jira using the required format: *As a [user role], I want [goal], So that [benefit]*. The individual story descriptions were entered directly into each Jira issue's Description field during backlog creation; the exact wording of each description was not captured in the supplied evidence screenshots, so it is not reproduced verbatim here. User-story descriptions were implemented in Jira using the required As a / I want / So that format, consistent with the Lab 1 functional requirements for each Epic.

5

Backlog Prioritization

Each User Story was assigned a priority level in Jira reflecting its immediate user or business impact:

- **High** — directly affects the rider or operator's immediate experience
- **Medium** — an important supporting capability that enables High-priority stories
- **Low** — lower immediate impact (not used in the current backlog)

Story	Priority	Rationale
Ingest GPS and Calculate ETA	Medium	Backend data pipeline — enables rider-facing features but is not itself user-visible.
View Live Bus ETA	High	Primary rider-facing feature; directly impacts commuter experience.
Estimate Bus Crowding	Medium	Adds value on top of core ETA tracking; depends on GPS ingestion first.
Monitor Fleet Status	High	Operationally critical for identifying fleet issues in real time.
Receive Transit Alerts	High	Time-sensitive commuter-facing notifications.
Handle Stale Data	Medium	Reliability/hardening work that supports the above, rather than new user value.

The priority values shown above are the actual values set on each Jira issue and were not altered for this report.

6
Story Point Estimation

Story points were assigned using the Fibonacci scale (0, 1, 2, 3, 5, 8, 13, …), which is used in Agile estimation because its non-linear spacing reflects growing uncertainty as task size increases. Story points represent relative **effort**, **complexity**, and **uncertainty** rather than exact time. Planning Poker was used as the estimation approach specified by the lab: each story was discussed briefly before a point value was agreed and recorded in Jira.

Story	Story Points
Ingest GPS and Calculate ETA	8
View Live Bus ETA	5
Estimate Bus Crowding	8
Monitor Fleet Status	5
Receive Transit Alerts	3
Handle Stale Data	5
Total	**34**

The two data/estimation-heavy stories — *Ingest GPS and Calculate ETA* and *Estimate Bus Crowding* — received the highest estimate (8 points) due to the complexity of processing live location data and deriving a crowding signal from it. Simpler, more self-contained features such as *Receive Transit Alerts* (3 points) required comparatively less effort and carried lower uncertainty.

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Sprint 1 — Core Tracking and Crowding

Story	Points
Ingest GPS and Calculate ETA	8
View Live Bus ETA	5
Estimate Bus Crowding	8
Sprint 1 Total	**21**

Sprint Goal: "Deliver the core real-time tracking, ETA visibility, and crowding-estimation capabilities."

Sprint 1 was created with a one-week duration and populated with the three highest-effort stories forming the foundation of the system, before Fleet Monitoring & Reliability work began in Sprint 2.

VERIFIED IN EVIDENCE

8
Sprint 1 Execution

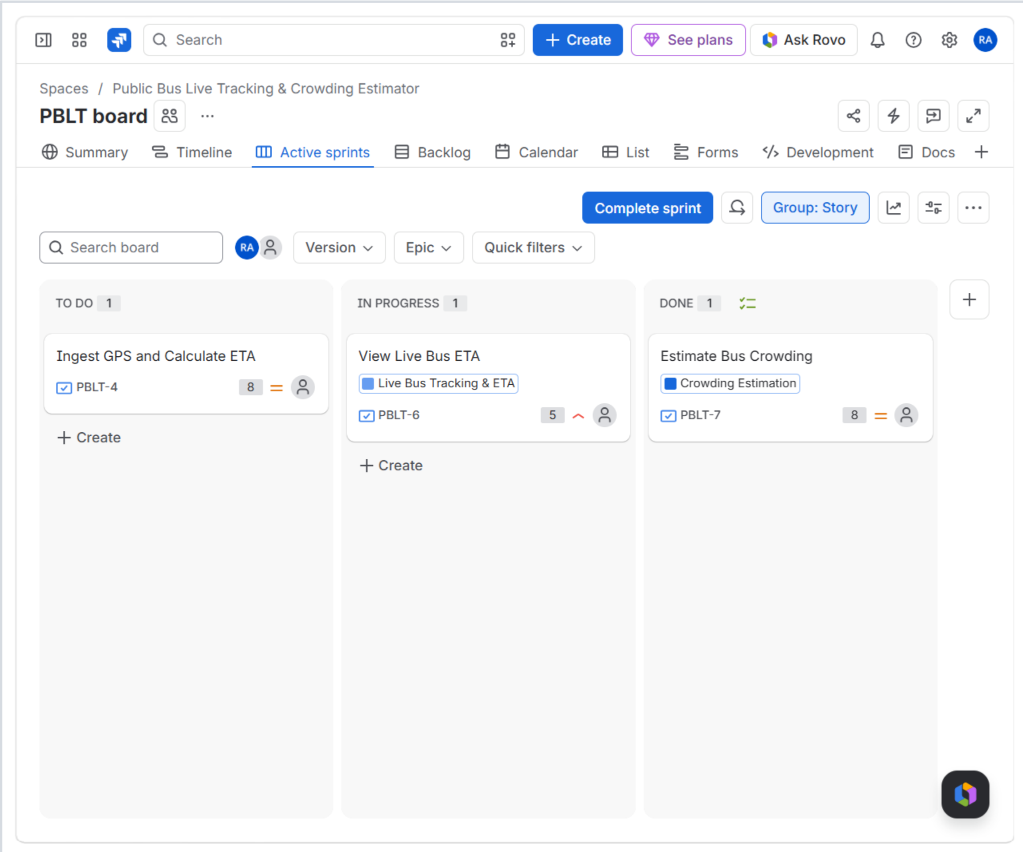

__Figure 2.__ Jira Active Sprint board captured mid-execution of Sprint 1, showing the three-column workflow in use.

The Scrum board demonstrates the required workflow:

To Do	In Progress	Done
Ingest GPS and Calculate ETA (8 pts) — planned, not yet started	View Live Bus ETA (5 pts) — actively being worked on	Estimate Bus Crowding (8 pts) — completed

This screenshot captures an intermediate state of Sprint 1 rather than its final state. Sprint 1's subsequent disappearance from the Backlog view (see Figure 3, Section 9) — with only Sprint 2 remaining listed — confirms that Sprint 1 was later completed via Jira's *Complete sprint* action, though no screenshot showing all three Sprint 1 stories simultaneously in the Done column was supplied.

PARTIALLY VERIFIED

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Sprint 2 — Fleet Reliability and Alerting

Story	Points
Monitor Fleet Status	5
Receive Transit Alerts	3
Handle Stale Data	5
Sprint 2 Total	**13**

Sprint Goal: "Improve operational visibility and reliability through fleet monitoring, alerts, and stale-data handling."

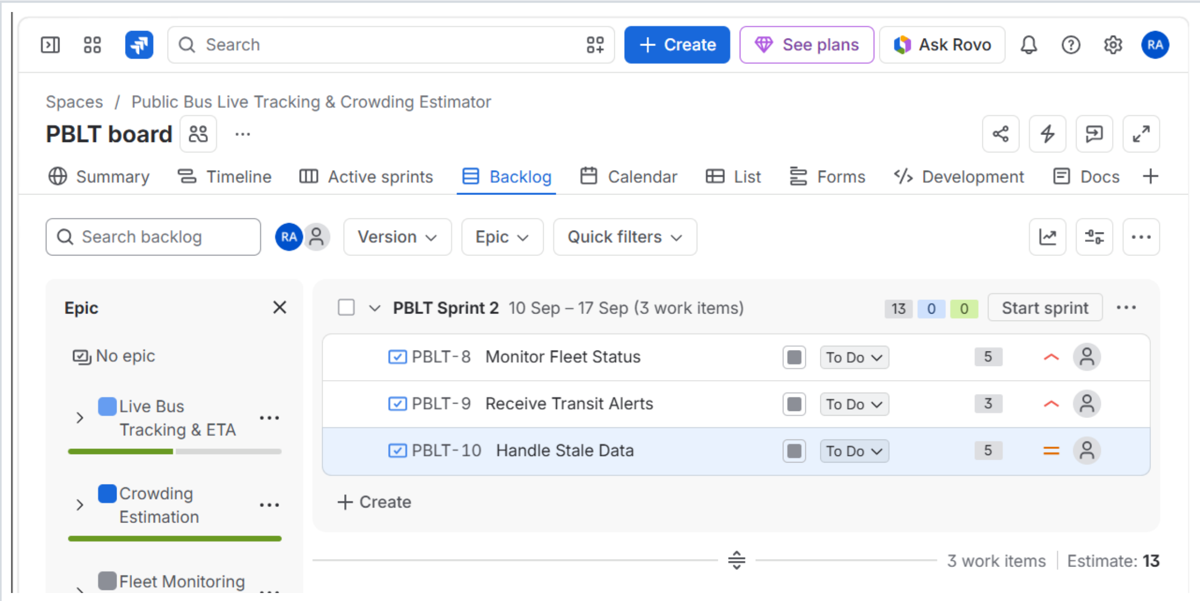

Figure 3. Jira Backlog view showing Sprint 2 planned with a one-week duration (10 Sep – 17 Sep) and its three assigned stories totalling 13 story points, prior to the sprint being started.

VERIFIED IN EVIDENCE

10

Sprint 2 Execution

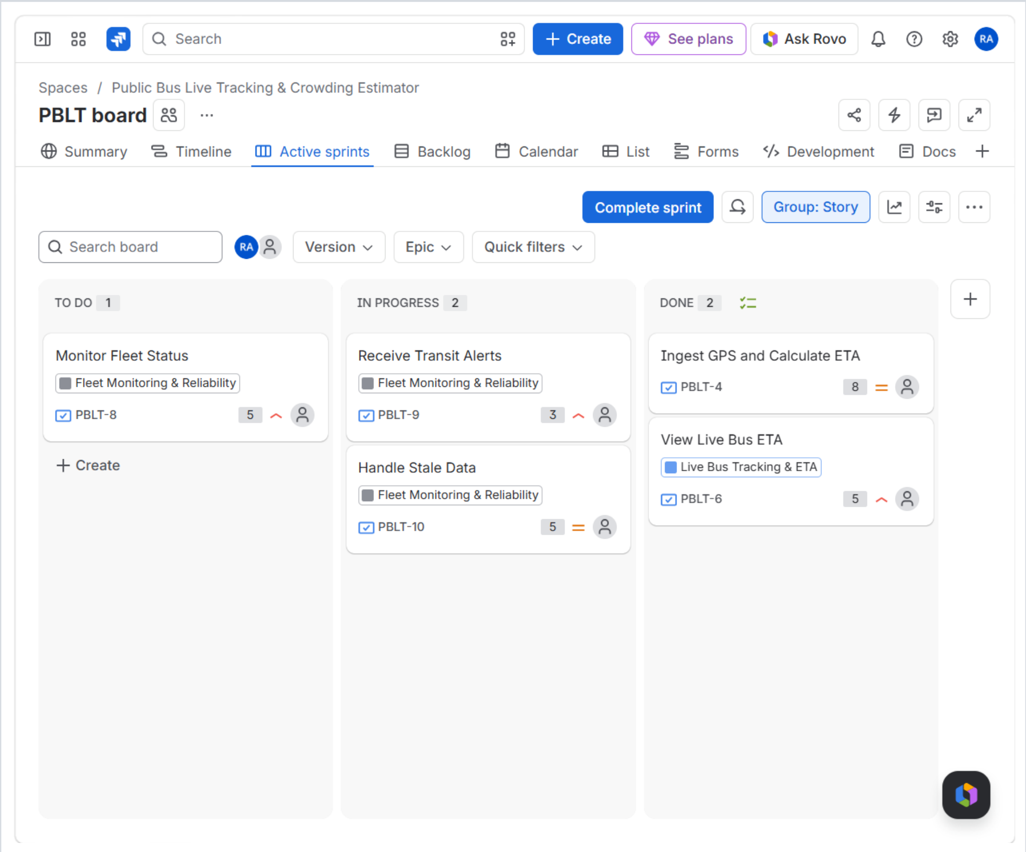

Figure 4. *Jira Active Sprint board for Sprint 2 at the time of evidence collection.*

The board shows **Monitor Fleet Status** (5 pts) in To Do, and **Receive Transit Alerts** (3 pts) and **Handle Stale Data** (5 pts) in In Progress. The Done column contains two items — however these are **Ingest GPS and Calculate ETA (PBLT-4)** and **View Live Bus ETA (PBLT-6)**, which are Sprint 1 stories, not the three stories scoped into Sprint 2.

Evidence Note: Based strictly on the supplied screenshot, none of the three Sprint 2 stories (Monitor Fleet Status, Receive Transit Alerts, Handle Stale Data) had reached Done at the time of capture. This should be verified and corrected in Jira — moving all three Sprint 2 stories to Done and capturing a fresh screenshot — before final submission, so the evidence clearly shows Sprint 2 completed.

NOT YET AVAILABLE

11

Story Points and Priorities — Jira Evidence

The table below is exported directly from the Jira Reports work-item table and shows the actual priority recorded against every issue in the PBLT project.

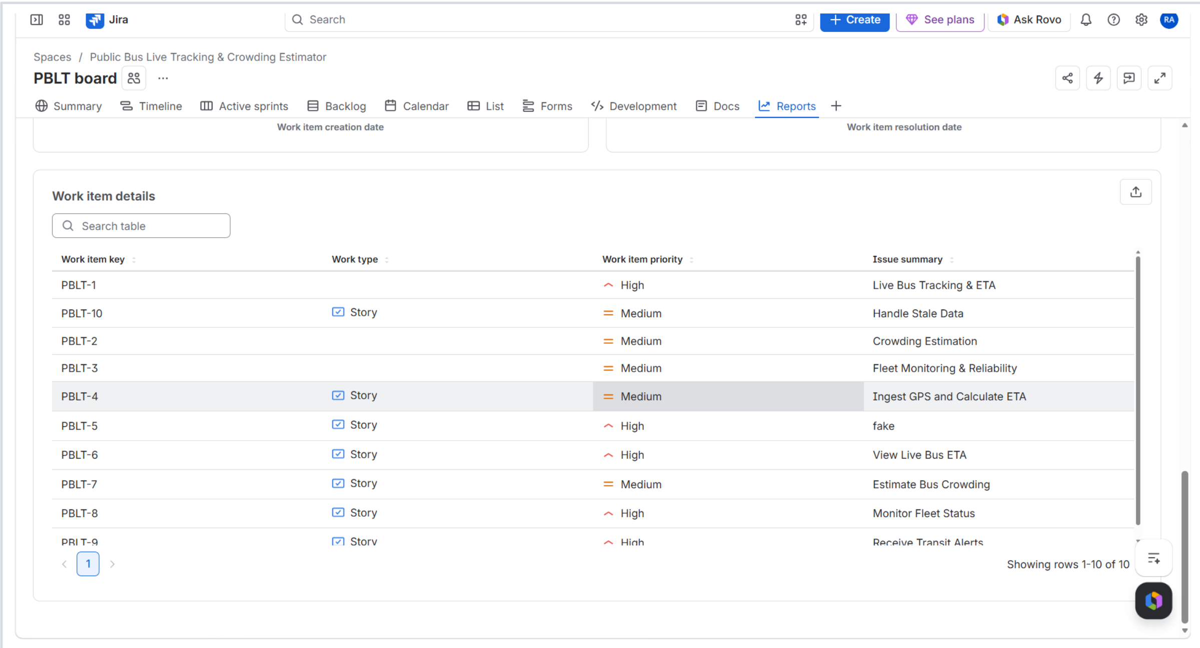

Figure 5. *Jira Reports > Work item details table listing all ten issues in the PBLT project with their type and priority.*

Evidence Note: This table also shows **PBLT-5 ("fake")**, a stray test issue created during setup that is not one of the six official User Stories and is not linked to any Epic or Sprint. It does not affect the 34-point backlog total or either sprint's scope, but it should be deleted from Jira before final submission for a clean project record.

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Burndown Chart Analysis

NOT YET AVAILABLE

A genuine Jira Burndown Chart screenshot (Reports > Burndown Chart, with the estimation statistic set to Story Points) was **not included** among the evidence supplied for this report. In line with the anti-fabrication requirement for this submission, no chart, remaining-work line, or numerical burndown values have been invented or estimated here.

To complete this deliverable:

1. Complete Sprint 2 in Jira (see Section 10) so both sprints have full history.
2. Navigate to the PBLT board > Reports > Burndown Chart.
3. Select the relevant sprint and set the estimation statistic to Story Points.
4. Capture a screenshot showing the axis labels, the guideline, and the remaining-values line.
5. Insert the screenshot here as Figure 6 and describe its actual shape in Section 13.

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Burndown Interpretation

This section is intentionally left as a template, to be completed once the genuine Burndown Chart screenshot (Section 12) is available. No values are assumed below.

Observation:

Describe exactly what the remaining-values line and guideline show once the chart is captured.

Interpretation:

State what the shape of the line indicates about how work was completed across the sprint.

Capacity:

State what this suggests about the team's ability to complete the sprint's planned 13 (or 21) story points.

Potential issue:

Note only if the chart visibly shows a flat period, late drop, or scope change.

Outcome:

State only what the chart supports — e.g. whether the sprint finished on, ahead of, or behind the guideline.

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Reflection

1. Did your estimations reflect the actual effort?

The Fibonacci scale helped distinguish clearly between heavier, data-processing-driven stories and lighter, self-contained ones. Both GPS ingestion and crowding estimation were rated 8 points, reflecting the complexity of working with live location data, while Receive Transit Alerts — a simpler notification feature — was rated only 3 points. Sprint 1's 21-point scope and Sprint 2's 13-point scope reflect this distinction between foundational and supporting work.

2. Was your backlog well-prioritized?

Largely yes. The three commuter-facing or operationally time-sensitive stories — View Live Bus ETA, Monitor Fleet Status, and Receive Transit Alerts — were marked High priority, while backend/supporting stories (Ingest GPS and Calculate ETA, Estimate Bus Crowding, Handle Stale Data) were marked Medium. This correctly separates what riders and operators directly interact with from the underlying infrastructure that enables it.

3. How did your simulated sprint align with your plan?

Sprint 1 (21 points) was planned and, based on the backlog view no longer listing it as active, appears to have been completed, delivering the core GPS/ETA/crowding functionality first. Sprint 2 (13 points) was planned with a correct one-week duration (10–17 Sep) and its execution began as intended; however, the supplied Sprint 2 board evidence shows that none of its three stories had yet reached Done at the time of capture, so this sprint's completion could not yet be verified.

4. What insights did the burndown chart give about your team's capacity?

No burndown chart screenshot was available in the supplied evidence for this submission, so no insight can be drawn from it at this time. This is flagged in Sections 12–13 as an outstanding item to capture once Sprint 2 is completed in Jira.

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Agile Learning Outcomes

✓ **Epic decomposition** — Functional requirements were grouped into three coherent Epics covering tracking, crowding, and fleet reliability.

✓ **User-story writing** — Six User Stories were written under those Epics using the As a / I want / So that convention.

✓ **Backlog prioritization** — Stories were ranked High or Medium based on direct user/operational impact.

✓ **Fibonacci estimation** — Story points (3, 5, 8) were assigned using the Fibonacci scale via a Planning Poker approach.

✓ **Sprint planning** — Two one-week sprints (21 and 13 points) were scoped from the prioritized backlog.

✓ **Sprint execution** — Stories were moved through the Scrum board's To Do / In Progress / Done columns.

✓ **Scrum workflow** — The PBLT board enforced the standard three-column Kanban-style workflow within each sprint.

✓ **Burndown analysis** — Identified as the outstanding step — to be completed once Sprint 2 finishes and the chart is captured.

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Conclusion

This lab demonstrated how Jira, as an Agile project-management platform, supports the full journey from raw functional requirements to a trackable, executable backlog. Requirements for the Public Bus Live Tracking & Crowding Estimator were transformed into three Epics and six User Stories, prioritized by business and user impact, and estimated using Fibonacci-based story points totalling 34. Two sprints — 21 points and 13 points respectively — were planned and executed through Jira's Scrum board, giving direct, first-hand experience of moving work through To Do, In Progress, and Done. The exercise reinforced why Agile teams favour small, estimable units of work: it makes both planning and progress tracking concrete rather than abstract. The main lesson from this submission is procedural as much as conceptual — Sprint 2's completion and the Burndown Chart still need to be finalized in Jira before this report can claim full closure of the sprint cycle, underscoring how sprint tracking is only useful when carried through to actual completion rather than left mid-flight.

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Final Evidence Checklist

Item	Status
3 Epics	✓
6 User Stories	✓
Epic–Story relationships	✓
Priorities (High/Medium)	✓
Fibonacci Story Points (total 34)	✓
Backlog organised by sprint	✓
Sprint 1 (planned & scoped)	✓
Sprint 1 completed	!
Sprint 2 (planned, 1-week duration)	✓
Sprint 2 completed (all 3 stories in Done)	✗
To Do → In Progress → Done demonstrated	✓
Burndown Chart	✗
Reflection questions answered	!
Conclusion	✓
Stray item PBLT-5 removed	✗

✓ = verified in supplied evidence ! = partially verified / needs a cleaner screenshot ✗ = not yet demonstrated in supplied evidence

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